

COMPUTATIONAL MACRO – PROBLEM SET 1

Antonela Gjinaj - Giovanni Manica

Due September 27, 2026

1

1.1

(a)

In this setting, the state variable, which we denote by x_t , is k_t , which summarizes all the previous states k_{t-j} , $j > 0$.

We move to identify the feasible set $\Gamma(x_t)$. We use the resource constraint

$$c_t + k_{t+1} = k_t^\alpha + (1 - \delta)k_t \iff k_{t+1} = k_t^\alpha + (1 - \delta)k_t - c_t$$

Using the feasibility constraints $k_{t+1} \geq 0$, $k_t \geq 0$, and $c_t > 0$, we find

$$\Gamma(x_t) = [0, k_t^\alpha + (1 - \delta)k_t)$$

Now, given k_t and k_{t+1} , we have $c_t = k_t^\alpha + (1 - \delta)k_t - k_{t+1}$. Since the one-period payoff is just utility from current consumption,

$$F(k_t, k_{t+1}) = u(c_t) = u(k_t^\alpha + (1 - \delta)k_t - k_{t+1})$$

In this setting, k_t is the right state variable because, once it is known, nothing else from the past will affect $\Gamma(x_t)$ or $F(k_t, k_{t+1})$.

If utility depended on c_{t-1} , k_t alone would no longer be sufficient, as there may be multiple histories leading to the same k_t but different c_{t-1} . So, the state would need to be augmented to $x_t = (k_t, k_{t-1})$ – or, equivalently, $x_t = (k_t, c_{t-1})$.

(b)

Plugging in our state, feasible set, and payoff function in the generic Bellman equation

$$v(x) = \max_{x' \in \Gamma(x)} \{F(x, x') + \beta v(x')\},$$

we get

$$\begin{aligned} v(k) &= \max_{k' \in [0, k^\alpha + (1-\delta)k]} \{F(k, k') + \beta v(k')\} \\ v(k) &= \max_{k' \in [0, k^\alpha + (1-\delta)k]} \{u(k^\alpha + (1-\delta)k - k') + \beta v(k')\} \end{aligned}$$

The $c > 0$ constraint never binds because of an Inada-type argument. We have in fact $u_c(c) = c^{-\gamma} \rightarrow \infty$ as $c \downarrow 0$. So, the optimum does not sit at $c = 0$. The $k' \geq 0$ constraint does not bind, either. As production is k^α with $0 < \alpha < 1$,

$$\frac{\partial k'^\alpha}{\partial k'} = \alpha k'^{\alpha-1} \rightarrow \infty \quad \text{as } k' \downarrow 0$$

because $\alpha - 1 < 0$. These two arguments let us conclude that the optimum is interior, and satisfies $0 < k' < k^\alpha + (1 - \delta)k$.

(c)

Starting from our Bellman equation

$$v(k) = \max_{k'} \{u(k^\alpha + (1 - \delta)k - k') + \beta v(k')\}, \quad [\text{where } k^\alpha + (1 - \delta)k - k' = c]$$

we take the FOC with respect to k' :

$$\frac{\partial v(k)}{\partial k'} : -\frac{\partial u(c)}{\partial k'} + \beta \frac{\partial v(k')}{\partial k'} = 0, \quad \text{or equivalently, } u_c(c) = \beta v_k(k')$$

To derive the envelope condition for this problem, we plug in $k' = g(k)$ in the Bellman equation and differentiate it with respect to k :

$$v(k) = u(k^\alpha + (1 - \delta)k - g(k)) + \beta v(g(k))$$

$$v_k(k) = u_c(c)[\alpha k^{\alpha-1} + (1 - \delta) - g_k(k)] + \beta g_k(k) v_k(g(k))$$

$$\Longleftrightarrow v_k(k) = u_c(c)[\alpha k^{\alpha-1} + (1 - \delta)] + g_k(k)[\beta v_k(g(k)) - u_c(c)]$$

From the previously derived FOC $u_c(c) = \beta v_k(k')$ and $k' = g(k)$, we have $\beta v_k(g(k)) - u_c(c) = 0$, and the previous expression becomes

$$v_k(k) = u_c(c)[\alpha k^{\alpha-1} + (1 - \delta)]$$

To arrive at the Euler equation, we evaluate it at one period ahead:

$$v_k(k') = u_c(c')[\alpha(k')^{\alpha-1} + 1 - \delta]$$

Plugging this into the FOC, we obtain

$$u_c(c) = \beta u_c(c')[\alpha(k')^{\alpha-1} + 1 - \delta]$$

which, defining $R(k') := \alpha(k')^{\alpha-1} + 1 - \delta$, results in the desired Euler equation

$$u_c(c) = \beta R(k') u_c(c')$$

(d)

The economic interpretation of the Euler equation $u_c(c) = \beta R(k') u_c(c')$ is that it equates the marginal utility of giving up one unit of consumption today, $u_c(c)$, with that of investing it as extra capital for tomorrow, $u_c(c')$, discounted by the discount factor β and multiplied by the gross rate of return $R(k')$, which counts in the marginal product of capital tomorrow, $\alpha(k')^{\alpha-1}$, and $(1 - \delta)$, that is, the share of that unit that survives after accounting for the depreciation rate δ .

1.2

(a)

For CRRA utility,

$$u(c) = \frac{c^{1-\gamma} - 1}{1 - \gamma}, \quad \gamma \neq 1,$$

while for $\gamma = 1$ we have $u(c) = \log c$. In either case, marginal utility can be written as

$$u_c(c) = c^{-\gamma}.$$

To obtain the inverse marginal utility function, let

$$x = u_c(c) = c^{-\gamma}.$$

Solving for c gives

$$c = x^{-1/\gamma},$$

and hence

$$u_c^{-1}(x) = x^{-1/\gamma}.$$

Consider now the term appearing in the definition of the Euler equation error:

$$u_c^{-1}(\beta R(k')u_c(c')).$$

Since $u_c(c') = (c')^{-\gamma}$, we obtain

$$\begin{aligned} u_c^{-1}(\beta R(k')u_c(c')) &= u_c^{-1}(\beta R(k')(c')^{-\gamma}) \\ &= [\beta R(k')(c')^{-\gamma}]^{-1/\gamma} \\ &= [\beta R(k')]^{-1/\gamma} c'. \end{aligned}$$

Plugging this back into the whole expression from (4),

$$\boxed{e(k) = 1 - (\beta R(k'))^{-1/\gamma} \frac{c'}{c}.$$

(b)

We are told that at state k_i , the policy is represented by the index g_i , so that tomorrow's capital is

$$k' = k_{g_i}.$$

Current consumption at grid point i is therefore

$$c_i = h(k_i) = k_i^\alpha + (1 - \delta)k_i - k_{g_i}.$$

Tomorrow's consumption is the consumption function evaluated at g_i , i.e.

$$c' = h(k') = h(k_{g_i}) = c_{g_i}.$$

The gross marginal return on capital evaluated at tomorrow's capital is

$$R(k') = \alpha(k')^{\alpha-1} + 1 - \delta = \alpha k_{g_i}^{\alpha-1} + 1 - \delta.$$

Therefore, evaluating the Euler equation error from part (a) at the grid point k_i gives

$$e_i := e(k_i) = 1 - \left[\beta \left(\alpha k_{g_i}^{\alpha-1} + 1 - \delta \right) \right]^{-1/\gamma} \frac{c_{g_i}}{c_i}, \quad i = 1, \dots, N.$$

Since the state tomorrow is k_{g_i} , applying the policy once more gives next-period capital $k_{g_{g_i}}$. Consequently,

$$c_{g_i} = k_{g_i}^\alpha + (1 - \delta)k_{g_i} - k_{g_{g_i}},$$

whereas c_i is just as defined above.

2